Designing a Rule-Accurate LAP Qualification Voice Agent

Approach Document | SalesAgents AI: AI Intern (Prompt Engineer) Assignment | **Upendra Rejeti**

1. Problem in one line

Build a voice agent for Home Credit that verifies an existing customer, presents a pre-approved Loan Against Property offer of up to 75 lakh rupees, collects 7 eligibility data points from natural, messy speech, and routes every call to exactly one correct outcome: **qualified handoff, disqualification, loan-transfer specialist, callback, or polite close**.

The hard part is not asking 7 questions. It is **reliably** remembering out-of-order answers, honouring corrections, stopping at the first disqualifier, spotting hidden transfer cases, and never handing off early, all while sounding like a human advisor.

2. Architecture: five layers expressed in natural language

Layer	What it does	Where in the prompt
1. Persona and voice	Identity, strict gender (including Hindi verb agreement), language lock, short speakable turns, AI honesty	Sections 1, 2, 4
2. Conversation flow	Greeting and identity gate, offer, checklist, recap, close	Section 6
3. State management	Silent 7-slot state rebuilt from history every turn; corrections overwrite	Section 5
4. Business rules	Per-item eligibility, immediate disqualification, transfer switch, 75 lakh rule, handoff gate	Sections 7, 8
5. Robustness	Diversions, RAG grounding, no-invention list, busy, wrong person, voicemail, silence, privacy	Sections 9, 10, 11

3. The core idea: an explicit per-turn reasoning loop

> **Extract → Validate → Update state → Route → Respond**
> An LLM has no memory outside the transcript, so the prompt tells the agent to *rebuild* its state from the full conversation history before every reply. This is what makes out-of-order answers and corrections reliable instead of accidental.

Routing priority (the first match wins): (1) busy or stop → callback; (2) existing loan on the property or EMI reduction → transfer; (3) a failed rule → disqualify; (4) a question → answer briefly and bridge back; (5) otherwise → ask the **first missing or unclear item** in order 1 to 7; (6) all valid → recap and handoff.

The First Missing Question rule satisfies two requirements that seem to conflict: capture items *sequentially*, yet never re-ask an item given *out of order*. The order is fixed, but already-filled slots are skipped. It also enforces the requirement to go back to items skipped during a diversion.

4. Eligibility decision table

#	Item	Eligible	Not eligible: immediate end	Clarify when
1	Property type	Residential, commercial, industrial, mixed use	Agricultural	"a plot", "land", "farmhouse"
2	Ownership	Sole or joint	None (never reject joint)	Property only in a relative's name: record it and continue

3	Original documents	Available (even if at home or in a locker)	Photocopies only, or lost	"probably somewhere"; "with the bank" → transfer
4	Loan amount	Up to and including 75 lakh	Over 75 lakh and refuses 75 lakh	Over 75 lakh → explain the limit and offer 75 lakh; a bare number → confirm the unit
5	Occupation + income	Salaried or self-employed, income via bank	Cash income (either occupation)	Mixed → "is the majority in the bank?"
6	Market value	Any estimate captured	None: no invented threshold	"Don't know" → ask once for a rough estimate
7	Tenure	3 to 15 years inclusive (months converted)	Under 3 or over 15 years, once confirmed	Out of range → confirm once (lets the customer self-correct)

5. Key design decisions and stated assumptions

Decision	Why
Identity is a hard gate: no offer details before the customer confirms	Privacy; the spec puts verification first. Third parties get no financial information.
"Original documents are with the bank" is a transfer trigger, not a disqualifier	It implies an existing mortgage on the same property, which is exactly the transfer case the spec describes.
A loan on a different property, or a personal or vehicle loan, is not a transfer trigger	The spec says "existing loan on the property". The agent clarifies once if unsure.
Transfer outranks disqualification when both come up together	Transfer is a different product with its own criteria, owned by the specialist.
One clarification before disqualifying on ambiguous or out-of-range answers	Avoids false rejections caused by speech-recognition errors or loose phrasing ("ten-twenty years"), while still ending immediately on clear failures.
Mixed income: decided by the majority mode	The spec only defines "bank" vs "cash". Majority-bank is the most defensible reading; it is documented so the business can change it.
No invented rules: market value, minimum loan, LTV, interest rates	The spec defines none of these, and inventing them is a top failure mode. Rates are always deferred to the senior expert.
RAG answers questions but never overrides the core rules	Keeps dynamic knowledge injection safe; when the RAG context is empty, the agent defers instead of guessing.
Short recap before handoff	Catches transcription errors and lets the customer correct them before a lead reaches a human expert, which improves lead quality.
Hindi in Devanagari with natural English banking words; gendered verb examples	Better TTS pronunciation and realistic Hinglish. "agent_gender must be adhered to strictly" matters most in Hindi grammar.
Voice-first output rules: 1 or 2 sentences, one question, no symbols, amounts in words	Voice UX: long or list-like turns sound robotic, and symbols like the rupee sign are mispronounced by TTS.
Explicit closing scripts plus the end_call tool	Guarantees the call actually ends after a terminal outcome, as the spec requires.

6. Requirement traceability

Assignment requirement	How it is satisfied	Test
Verify the customer; busy → callback time	Stage A gate, busy branch, relative time resolved from current_date and current_day	T12, T23
Offer framed as a loyalty reward, up to 75 lakh	Stage B scripts (English and Hindi)	T1
7 items sequential; out-of-order answers not re-asked	State plus the First Missing Question rule	T11, T13
Immediate disqualification	Section 8.1 plus routing priority 3	T2 to T6, T14
Transfer logic switch	Section 8.2, including hidden triggers	T9, T10, T18, T19
Over 75 lakh → explain and offer the maximum	Section 7.4	T7, T7b

Handoff only after all items; go back to skipped ones	Section 8.3 gate	T15, T20
All 10 prompt variables	Section 0, fallbacks, gender and language rules, RAG rules	T16
Natural speech, interruptions, fillers; not a yes/no bot	Sections 4, 7 and 9: open questions, synonyms, corrections	T13, T14, T21

7. Testing methodology

24 scripted scenarios in three tiers: **core business flow** (T1 to T12), **adversarial and natural speech** (T13 to T24: contradictions, all-at-once answers, interruptions, hidden transfer triggers, early-handoff pressure, scam checks, Hindi with both genders), and a **results log** recording the expected outcome, actual outcome, pass or fail, and any fix applied. Retell post-call analysis fields extract the outcome and all 7 slots from each call, plus an *any_question_repeated* quality flag, so the results can be measured rather than judged by impression.

8. Possible production improvements

- Move slot state out of the prompt into a **structured function call** (for example `update_lead(slot, value)`) so a deterministic layer enforces the gates and writes leads to the CRM.
- A **conversation-flow agent** (node-based) for hard gates, with the LLM handling language understanding inside each node.
- Automated regression: simulated-customer LLM test runs over all 24 scenarios after every prompt change.
- Callback scheduling through a calendar or CRM API; warm transfer to a live expert when one is available.
- DND and consent checks before dialling; call-time windows per regulatory guidelines.